PS 160 — Physics II – Test #1

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Date: Thursday 10 October 2024

Name _____

Score: [ ]/150

- RELAX.
- By starting this exam, you commit yourself to Embry-Riddle Aeronautical University's Code of Conduct.
- Look over all questions before you begin, and attempt the easier ones first. Place your answers in the boxes provided.
- To receive full credit you must SHOW your work. Your handwritten work MUST be submitted in a clear, concise manner.
- Partial credit will be based on the quality and correctness of your work and your final answer. Correct answers with no work receive zero credit. Correct answer with partially correct work will receive partial credit.

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knowledge questions

[30 points]

Fluids

1. [**2 points**] Write the definitions of density and pressure.

$$\rho = \frac{m}{V}$$

$$P = \frac{F_\perp}{A}$$

2. [**2 points**] Write the hydrostatic condition and its solution for constant density.

$$\frac{dp}{dy} = -\rho g$$

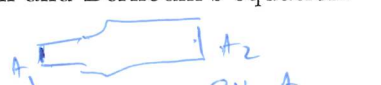
$$dp = -\rho g\, dy$$

$$\int_{p_1}^{p_2} dp = -\int_{y_1}^{y_2} \rho g\, dy = -\rho g \int_{y_1}^{y_2} dy = \rho g h \quad : h = y_1 - y_2.$$

$$P_2 = P_1 + \rho g h$$

3. [**2 points**] Write the continuity equation and Bernoulli's equation.

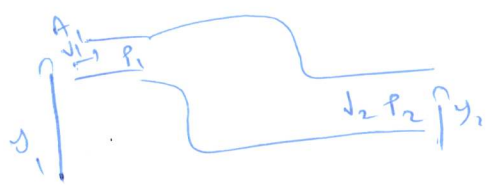
$$\boxed{\rho v A = \text{constant}}$$



$$\rho v_1 A_1 = \rho v_2 A_2$$

$$\boxed{v_1 A_1 = A_2 v_2}$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = \text{constant} = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$



Oscillations

4. [**2 points**] Write the differential equation for simple harmonic motion and its solution.

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \qquad x(t) = A\cos(\omega t + \varphi)$$

5. [**3 points**] Write the formulas for the angular frequency of small oscillations of a mass-spring system, a simple pendulum, and a physical pendulum.

$$\omega_0 = \sqrt{\frac{k}{m}} \quad : \text{mass-spring system}$$

$$\omega_0 = \sqrt{\frac{g}{L}} \quad : \text{simple pendulum}$$

$$\omega_0 = \sqrt{\frac{mgd}{I}} \quad : \text{physical pendulum}$$

6. **[3 points]** Write the relations between the frequency, angular frequency, and period.

$f = \frac{1}{T}$

$\omega = 2\pi f$

$\omega = \frac{2\pi}{T}$

Waves and Sound

(a) **[3 points]** Write the wave function for a right-going wave, and the definitions of wavenumber and wave speed.

$y(x,t) = A\cos(kx - \omega t + \varphi)$

$k = \frac{2\pi}{\lambda}$ $\quad v = \lambda f = \frac{\omega}{k}$

(b) **[3 points]** Write the formulas for the speed of waves on a string, in a solid, and in a fluid.

$v = \sqrt{\frac{F}{\mu}}$ $\quad \mu = \frac{m}{L}$

$v = \sqrt{\frac{Y}{\rho}}$ $\quad v = \sqrt{\frac{B}{\rho}}$

(c) **[2 points]** Write the formulas for the frequencies and wavelengths of standing waves.

string: $f_n = n\frac{v}{2L}$ $\quad \lambda_n f_n = v \Rightarrow \lambda_n = \frac{v}{f_n}$

$\lambda_n = \frac{2L}{n}$

Pipe: open at both ends. $f_n = n\frac{v}{2L}$

open at one end $f_n = n\frac{v}{4L}$

(d) **[2 points]** Write the formulas for the average power of waves on a string and the intensity of sound waves.

$P_{av} = \frac{1}{2}\sqrt{F\mu}\, A^2\omega^2$: string

$I_{av} = \frac{1}{2}\sqrt{B\rho}\, A^2\omega^2$

(e) **[2 points]** Write the formulas for the pressure amplitude of sound waves and the intensity-distance relationship.

$P_{max} = AKB$

$I_1 r_1^2 = I_2 r_2^2$

(f) **[2 points]** Write the formulas for the sound intensity level and the reference intensity.

$$\beta = 10 \log\left(\frac{I}{I_0}\right)$$

$$I_0 = 10^{-12} \frac{W}{m^2}$$

(g) **[2 points]** Write the formulas for the Doppler shifted frequency (source to the right of listener) and the beat frequency.

$$f' = f_s \frac{v \pm v_L}{v \pm v_s}$$

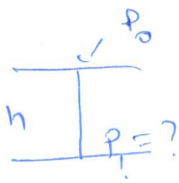
$$f_b = f_2 - f_1$$

Calculation questions

[120 points]

1. Calculate the pressure at the bottom of a cylinder of mercury 47 cm deep if the ambient atmospheric pressure is 105,368 Pa. The density of mercury is 13,600 kg/m^3.

$p =$ [ ] Pa



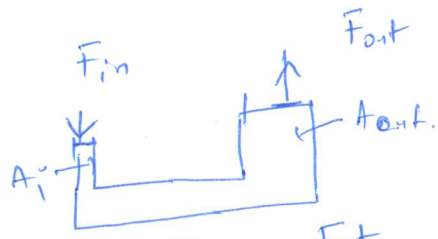
$P_1 = P_0 + \rho g h$

$= 105{,}368 + (13{,}600)(9.8)(47 \times 10^{-2})$

2. A hydraulic jack has input piston with radius 0.17 m and output piston with radius 1.48 m. If the input force is 413 N;

(a) calculate the output force in N.

$F_{out} =$ [ ] N



$\frac{F_{in}}{A_{in}} = \frac{F_{out}}{A_{out}}$

$A_{in} = \pi r_{in}^2$

$A_{out} = \pi r_{out}^2$

$F_{out} = \frac{F_{in}}{A_{in}} \times A_{out}$

$F_{out} = F_{in} \cdot \frac{\pi r_{out}^2}{\pi r_{in}^2} = F_{in}\left(\frac{r_{out}}{r_{in}}\right)^2$

$F_{out} = (413)\left(\frac{1.48}{0.17}\right)^2$

(b) Determine the pressure applied to the fluid in the lift.

$p =$ [ ] Pa

$P = \frac{F_{in}}{A_{in}} = \frac{F_{out}}{A_{out}}$

$P = \frac{413}{\pi \cdot (0.17)^2}$

3. A hose with radius 1.9 cm fills a rectangular pool that is 8 m in width, 3 m in width, and 3 m in depth in 3 hours. At what speed does water come from the hose?

$v =$ _____ m/s

Volume Speed

$$\text{Flow} = \frac{\Delta V}{\Delta t} = A v$$

$$v = \left(\frac{\Delta V}{\Delta t}\right) / A$$

$$V = (3)(3)(8)$$

$$3h = (3)(60)(60)$$

$$= (3)(3600)$$

$$A = \pi r^2$$

$$= \pi \cdot (1.9 \times 10^{-2})^2$$

4. Suppose a horizontal block-spring has a spring constant k=624 N/m and a block mass m=4.4 kg. Calculate the period of s of its harmonic motion.

$T =$ _____ s

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{k}{m}}} = 2\pi\sqrt{\frac{m}{k}}$$

$$= 2\pi\sqrt{\frac{4.4}{624}}$$

5. A 0.25 kg ideal harmonic oscillator has a total mechanical energy of 4.0 J. If the oscillation amplitude is 20.0 cm, what is the oscillation angular frequency?

$\omega =$ _____ rad/s

$$\boxed{E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kA^2} = \frac{1}{2}mv_{max}^2$$

$$E = \frac{1}{2}kA^2$$

$$k = \frac{2E}{A^2}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{2E}{mA^2}} = \sqrt{\frac{2(4)}{(0.25)(20 \times 10^{-2})}}$$

6. A simple pendulum has length 0.67 m and mass 6.9 kg. If the local acceleration of gravity on this planet is 5.9 m/s^2, calculate the frequency of the motion.

$f =$ [ ] Hz

$\omega = \sqrt{\frac{g}{L}}$

$\omega = 2\pi f$

$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{g}{L}} = \frac{1}{2\pi}\sqrt{\frac{5.9}{0.67}}$

7. Suppose a wave function is given by $y(x,t) = 5\cos(3t + 0.5x - 2.1)$ where all values are in MKS units. Then ...

(a) the amplitude is [5] m

(b) the angular frequency is [3] rad/s

(c) the wavenumber is [0.5] rad/m

(d) the phase shift is [-2.1] rad

(e) the frequency is [$\frac{3}{2\pi}$] Hz

$f = \frac{\omega}{2\pi}$ $(\omega = 2\pi f)$

(f) the period is [$\frac{2\pi}{3}$] s

$T = \frac{1}{f}$

(g) the wavelength is [$\lambda = \frac{2\pi}{0.5}$] m

$k = \frac{2\pi}{\lambda} \Rightarrow \cancel{k = \frac{2\pi}{\lambda}}$ $\frac{2\pi}{k}$

$v = \lambda f$

(h) the propagation speed of the wave is [$\left(\frac{2\pi}{0.5}\right)\left(\frac{3}{2\pi}\right) = 6$] m/s

(i) the transverse **speed** of the wave at (0, 0) is [ ] m/s

$\left(\frac{\partial y}{\partial t}\right)_{(0,0)} = -(5)(3)\sin(3\underset{0}{t} + 0.5\underset{0}{x} - 2.1)$

$= -(5)(3)\sin(-2.1)$

$v_t(0,0) = 15\sin(2.1)$ rad

$\times 10^{-3}$

8. A 2.00-m long piano wire with a mass per unit length of 12.0 g/m is under a tension of 8,000 N. What is the frequency of the second harmonic fundamental mode of vibration of this wire?

$f =$ _____ Hz

$\mu = \frac{kg}{m}$

$f_n = n \frac{v}{2L}$

$n = 2$

$\mu = \frac{m}{L}$

$v = \sqrt{\frac{F}{\mu}}$

$= \sqrt{\frac{FL}{m}} = \sqrt{\frac{8000(2)}{}}$

$f_2 = \frac{(2)\sqrt{\frac{8000}{(12)\times 10^{-3}}}}{2 \times (2)}$

9. A sound wave in an unknown fluid medium takes 11 ms to travel 20.8 m. What is the bulk modulus of the fluid in GPa, if the density is 1,229 kg/m^3? (1 GPa = 10^9 Pa)

$B =$ _____ GPa

$v = \frac{d}{t} = \frac{20.8}{(11)(10^{-3})}$

$v = \sqrt{\frac{B}{\rho}}$

$v^2 = \frac{B}{\rho}$

$B = \rho v^2$

10. A siren is designed to create a decibel level of 119 dB at a distance of 10 m. Calculate the sound intensity in W/m^2 at the same distance if there are 4 sirens all sounding at that same location. ($I_0 = 10^{-12}$ W/m^2)

β?

$I =$ _____ W/m^2

$I_N = N I_1$

$N = 4$

$I_4 = 4 I_1$

$\beta_1 = 10 \log\left(\frac{I_1}{I_0}\right)$

$\frac{\beta_1}{10} = \log\left(\frac{I_1}{I_0}\right)$

$10^{\frac{\beta_1}{10}} = \frac{I_1}{I_0}$

$\Rightarrow I_1 = I_0 \, 10^{\frac{\beta_1}{10}}$

$I_4 = 4 I_0 \, 10^{\frac{\beta_1}{10}}$

$1\,ms = 10^{-3}\,s$

11. Two guitars that are out of tune with each other play a note that has a period of 1.860 ms on one guitar and 1.889 ms on the other. What is the beat frequency (in Hz) heard by the two guitarists when they play the notes together? Assume that the speed of sound is 344 m/s.

$\Delta f =$ [ ] Hz

$f_b = f_2 - f_1$

$f = \frac{1}{T} \Rightarrow \cancel{f_b = \frac{1}{T}}$ $\boxed{f_b = \left|\frac{1}{T_2} - \frac{1}{T_1}\right|}$

12. A motorcyclist is traveling at 49 m/s away from a police car at rest. If the policeman blows his horn at a frequency of 439 Hz, what frequency in Hz does the motorcyclist hear? The speed of sound is 340 m/s.

$f =$ [ ] Hz

$49\,\frac{m}{s}$ $V_s = 0$

$f = f_s \frac{V \pm V_L}{V \pm \cancel{V_s}}$

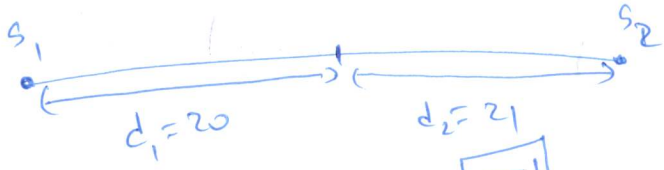
$f = f_s\left(\frac{V - V_L}{V}\right)$

$= (439)\left(\frac{340 - 49}{340}\right)$

Bonus

The sound from a single source can reach point O by two different paths. One path is 20.0 m long and the second path is 21.0 m long. The sound constructively interferes at point O. What is the minimum frequency of the source if the speed of sound is 340 m/s?

$f =$ _____ Hz



$\boxed{m = 1}$

$|d_2 - d_1| = \delta = m\lambda$

$d_2 - d_1 = 21 - 20 = 1\ m$

$\lambda_{min} =$

$\lambda_{min} =$

$\boxed{\lambda_{min} = 1\ m}$

$\lambda \cdot f = v$

$f = \frac{v}{\lambda} = \frac{340}{1} = 340\ Hz$

The End